

Does Female Unemployment Contribute to Changing Fertility Patterns in India?

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14 December 2024

Abstract

This study explores the impact of female unemployment on fertility rates in India, with urbanization as a control variable, using time series data from 1991 to 2023. The primary research question examines whether female unemployment negatively influences fertility rates in India over this period. Employing ordinary least squares (OLS) regression analysis, the study finds a significant negative relationship between both female unemployment and fertility rates, as well as urbanization and fertility. The results suggest that rising female unemployment and increased urbanization are both associated with lower fertility rates in India. These findings highlight the need for policy interventions that address the economic and social challenges faced by women in the labor market and the implications of urbanization on family planning decisions. The study contributes to the understanding of demographic trends in India and provides directions for future research on the broader socio-economic factors affecting fertility.

1 Introduction

Fertility plays a crucial role in shaping demographic trends and determining population growth, which has significant implications for a country's economic development, social structures, and healthcare systems. In particular, fertility rates reflect underlying socio-economic factors, including women's empowerment, access to education, and labor market participation. High or low fertility rates can impact a nation's workforce, dependency ratios, and long-term economic sustainability. For India, a country with a vast and diverse population, understanding the factors influencing fertility decisions is vital for designing effective policies related to family planning, healthcare, and economic development. Over the past several decades, India has experienced a gradual decline in fertility rates, driven by a combination of factors such as improved healthcare, increased female education, and changing social norms. From a high fertility rate in the 1990s, India has seen a steady reduction in fertility, signaling a transition toward lower population growth. This decline is particularly significant in the context of India's ongoing socio-economic transformation, where shifts in

labor force participation and urbanization are reshaping family dynamics and childbearing choices. One of the key factors influencing fertility in India is female unemployment. Women's participation in the labor force is influenced by a variety of economic and social factors, and the rate of female unemployment has been a subject of growing concern in recent years. Female unemployment is an important determinant because it reflects the barriers women face in accessing economic opportunities and their broader role in economic development. While some studies suggest that female unemployment may have a negative impact on fertility—by reducing economic stability and delaying marriage or childbearing—other studies have proposed that higher unemployment may reduce the cost of raising children and, consequently, increase fertility rates. In addition to female unemployment, urbanization plays a significant role in shaping fertility decisions. As India becomes increasingly urbanized, living conditions, access to education, healthcare, and employment opportunities in urban areas influence family planning choices. Urban areas generally offer better economic opportunities for women but may also present challenges, such as higher costs of living, which could lead to lower fertility rates.

2 Literature review

The purpose of this study is to explore how female unemployment affects fertility rates, focusing on the Indian context. Fertility and its determinants have been the subject of extensive research globally, with various studies presenting conflicting evidence regarding the relationship between unemployment and fertility. While some researchers argue that female unemployment increases fertility rates due to reduced opportunity costs, others highlight negative associations driven by economic insecurity. This section reviews key studies that form the theoretical and empirical foundation of this research, while identifying gaps that this study seeks to address.

Research on the relationship between female unemployment and fertility has produced mixed results. Some studies argue that higher female unemployment leads to increased fertility, while others report the opposite. • In the paper *The Effects of Unemployment on Fertility: Evidence from England published in the British Journal of Economics and Applied Economics*, **Cevat Giray Aksoy** demonstrates that female unemployment tends to increase births. The study provides evidence that women are more likely to have children during periods of unemployment due to reduced opportunity costs and fewer career-related constraints. • Supporting this perspective, **Debraj Ray**, in his book *Development Economics*, emphasizes that higher unemployment rates lower the opportunity cost of having children, thereby encouraging higher fertility rates. This observation is particularly relevant in societies where alternative economic opportunities for women are limited. Conversely, other studies find a negative association between unemployment and fertility: • The paper *Unemployment and Fertility: The Relationship Between Individual and Aggregated Unemployment and Fertility During 1994-2014 in Norway* by **Axel Peter Kristensen**

and **Trude Lappegård** finds a negative association between individual unemployment and birth rates. It argues that financial insecurity during periods of unemployment discourages couples from having children, as they prioritize economic stability over family expansion. • Similarly, in *Not the Right Time for Children*, **Flavia Cavallini** finds that an increase in unemployment rates leads to a rise in abortion rates, suggesting that economic instability often results in delaying or avoiding childbearing altogether.

The relationship between women’s employment and fertility is also a critical consideration. Employment opportunities for women are often associated with lower fertility rates, as balancing work and family responsibilities becomes challenging. • The paper *Women’s Employment and Fertility in a Global Perspective* by **Julia Behrman and Pilar Gonalons-Pons** finds that employed women tend to have fewer children, as career aspirations and workplace demands often take precedence over family expansion. This observation underscores the competing priorities faced by women in balancing professional and personal life.

Urbanization, a significant socio-economic factor, has been widely linked to declining fertility rates. The transition from rural to urban living often introduces lifestyle changes, increased access to education, and better healthcare, all of which contribute to fertility decline. • Studies by **Mathias Lerh** highlight the persistent decline in fertility in urban areas, where access to modern contraception and education is higher, and traditional family norms are weaker. Urbanization reshapes societal expectations, placing greater emphasis on economic opportunities and smaller family sizes.

While the above studies provide valuable insights, most of the research is focused on Western countries, with limited exploration of the Indian context. India presents a unique socio-economic and cultural environment where female unemployment, urbanization, and fertility interact in complex ways. • Despite the richness of global literature, there is a lack of studies examining the impact of female unemployment on fertility within a time-series framework for India. Most existing research focuses on cross-sectional analyses or regional comparisons, leaving a gap in understanding how these relationships evolve over time in India. • The dynamics of female unemployment in India exhibit significant transitions, with unemployment rates remaining stable until 1994, increasing steadily until 2005, and then gradually declining until 2018, followed by a sharp fall thereafter. This trajectory presents an opportunity to study the effects of female unemployment on fertility rates longitudinally.

The reviewed studies provide contrasting evidence on the relationship between female unemployment and fertility, influenced by socio-economic contexts. While some studies suggest that unemployment encourages higher fertility due to reduced opportunity costs, others highlight the negative impact of financial insecurity on childbearing decisions. The role of urbanization in lowering fertility rates is also well-documented, emphasizing the importance of considering it as a control variable in fertility analyses. However, these studies often overlook the country-specific context of India, particularly the evolution of these variables over time. This research seeks to address this gap by analyzing the relationship between female unemployment and fertility rates in India from 1991 to 2023,

using urbanization as a control variable in a time-series setup. By doing so, it aims to provide valuable insights into the complex interplay of these factors in shaping fertility trends in India.

3 Methodology

3.1 Data Source

This study utilizes secondary data from the *World Bank Development Indicators*, covering the period **1991 to 2023**. The dataset includes time-series data for fertility rates, female unemployment rates, and urbanization levels in India. These variables are selected to understand the interplay between female unemployment and fertility, with urbanization incorporated as a control variable to account for socio-economic changes.

3.2 Variables

- **Dependent Variable:** Fertility rate (total births per woman), which reflects reproductive behavior in India over time.
- **Independent Variable:** Female unemployment rate, representing the percentage of unemployed women in the labor force.
- **Control Variable:** Urbanization, measured as the proportion of the population living in urban areas. Urbanization captures structural transformations in society, such as access to education, healthcare, and employment opportunities, that may influence fertility behavior.

3.3 Analytical Approach

The relationship between female unemployment and fertility rates is examined using *Ordinary Least Squares (OLS) regression*. The time-series nature of the data enables a dynamic analysis of how these relationships evolve over the 32-year study period. The regression equation is defined as:

$$Fertility_t = \beta_0 + \beta_1 \cdot Unemployment_{f,t} + \beta_2 \cdot Urbanization_t + \epsilon_t$$

Where:

- $Fertility_t$: Fertility rate at time t
- $Unemployment_{f,t}$: Female unemployment rate at time t
- $Urbanization_t$: Urbanization level at time t
- ϵ_t : Error term

3.4 Justification of Method

OLS regression is an appropriate method to estimate linear relationships between variables. It enables an analysis of how female unemployment impacts fertility while accounting for urbanization as a control variable. Time-series data adds depth by capturing variations across the study period, providing insights into temporal trends.

3.5 Data Preparation

Data was examined for missing values and inconsistencies. Observations with incomplete data were excluded to ensure robust and reliable analysis. Tests for stationarity and other OLS assumptions were conducted before performing the regression.

This methodological framework offers a comprehensive approach to exploring the impact of female unemployment on fertility rates in India over time.

4 Result and Analysis

The Ordinary Least Squares (OLS) regression model was employed to examine the relationship between fertility rates (dependent variable), female unemployment rates (independent variable), and urbanization (control variable) using time-series data for India from 1991 to 2023. The results from the regression analysis are presented below.

The estimated intercept in the model is 9.7005, with a highly significant p-value of less than 0.001. This indicates that when both female unemployment and urbanization are zero, the baseline fertility rate is approximately 9.7 births per woman. Although the intercept itself does not carry direct substantive meaning, it provides a reference point for the model's estimates.

The coefficient for female unemployment is -0.0896 with a p-value less than 0.001, indicating a statistically significant negative relationship between female unemployment and fertility rates. Specifically, this means that for each one-percentage-point increase in female unemployment, the fertility rate decreases by approximately 0.0896 births per woman, while controlling for urbanization. This negative relationship suggests that as unemployment among women rises, fertility rates in India tend to decline. This finding contrasts with some studies that have found positive effects, where higher unemployment is associated with increased fertility, possibly due to lower opportunity costs of having children. In the Indian context, this negative relationship may be influenced by factors such as economic insecurity, changing social norms, or greater access to family planning and reproductive health services.

Similarly, the coefficient for urbanization is -0.2034, which is also statistically significant (p-value < 0.001). This indicates a negative association between urbanization and fertility rates, with each percentage-point increase in urbanization being associated with a decrease in fertility rates by 0.2034 births per

woman. This result supports existing literature, which suggests that urbanization is often linked to lower fertility due to the availability of better education, employment opportunities, and healthcare services, particularly for women. Urbanization may also lead to shifts in family structures and aspirations, further reducing fertility rates.

The model’s fit is strong, with a Residual Standard Error (RSE) of 0.09654, indicating the average deviation of observed fertility rates from the predicted values. The Multiple R-squared value is 0.978, meaning that the model explains approximately 97.8

In conclusion, the regression analysis reveals that both female unemployment and urbanization have a significant negative impact on fertility rates in India. These findings are consistent with the broader socio-economic changes occurring in the country, including increased female participation in the workforce, greater urban migration, and evolving family dynamics. Further research is needed to better understand the underlying factors driving these trends and their implications for future fertility patterns in India.

Regression Results Table

Below is the regression output table that summarizes the key findings:

Variable	Estimate	Std. Error	t-value	Pr(> t)
(Intercept)	9.7005	0.2319	41.827	1.22e-16
unemployment_f	-0.0896	0.0149	-6.033	1.27e-06
urbanization	-0.2034	0.0056	-36.097	1.22e-16

Table 1: OLS Regression Results for Fertility Rate, Female Unemployment, and Urbanization

5 Conclusion

This research investigates the effect of female unemployment on fertility rates in India for the period 1991–2023, with urbanization as one of the control variables. The results of the OLS regression analysis suggest that indeed, there is round about a negative relationship between female unemployment and fertility rate which reverses the traditional view of female unemployment and fertility rate being positively correlated. Specifically, In the Indian context, there is an inverse correlation between female unemployment and prevalence of child birth. Thus, the socio-economic conditions of India, wherein a high degree of financial instability and uncertainty is prevalent, contribute to the negative attitude towards child-bearing. Consequently, there is also a positive correlation between urbanization and the length of fertility period, but an inverse relationship with the number of children born. This indicates how significantly urbanization has altered demographic structures by providing more education, better health care, more jobs, change in family structure, and family ideals. All these factors are responsible for the large decrease in fertility rate over the last three decades in

India. The study adds to the literature in that it has conducted a time-series analysis within an Indian context which is a gap in the literature as highlighted in earlier researches. However, whereas most of the literature from different parts of the world finds it difficult to extract a consistent relationship between unemployment and fertility, this research simply illustrates that country

6 References

1. Aksoy, C. G. (2019). *The Effects of Unemployment on Fertility: Evidence from England*. *British Journal of Economics and Applied Economics*.
2. Behrman, J., & Gonalons Pons, P. (2020). *Women's Employment and Fertility in a Global Perspective*. *Population and Development Review*, 46(3), 469–491. <https://doi.org/10.1111/padr.12378>
3. Cavallini, F. (2019). *Not the Right Time for Children: The Effect of Unemployment on Fertility and Abortion*. *Social Science Research*, 74, 1–17. <https://doi.org/10.1016/j.ssresearch.2018.08.002>
4. Kristensen, A. P., & Lappegard, T. (2018). *Unemployment and Fertility: The Relationship Between Individual and Aggregate Unemployment and Fertility in Norway, 1994-2014*. *European Journal of Population*, 34(2), 175–202. <https://doi.org/10.1007/s10680-018-9469-4>
5. Lerh, M. (2016). *Urbanization and Its Impact on Fertility Rates: Evidence from Europe*. *Population Studies*, 70(1), 56–74. <https://doi.org/10.1080/00324728.2015.1075061>
6. Ray, D. (1998). *Development Economics*. Princeton University Press.
7. World Bank. (2024). *World Development Indicators*. World Bank. Retrieved from <https://data.worldbank.org/>
8. Zhang, W., & Liu, L. (2021). *The Role of Female Employment in Fertility Decline: A Study on Urbanization and Labor Force Participation*. *Asian Economic Policy Review*, 16(3), 103–118. <https://doi.org/10.1111/aepr.12335>